

Submission Deadline: 20th July 2026

DDL and DML

- Upload all your necessary documents as a single PDF file in VTOP [Mandatory] and Moodle [optional] on or before the deadline.
- No other form of submission will be acceptable.
- If you failed to upload in VTOP on or before the deadline, but successfully uploaded in Moodle, then 2 marks of penalty will be imposed on the secured marks.
- If you fail to upload your assignment in both VTOP and Moodle, then your assignment will not be evaluated and ZERO (0) mark will be awarded.

Write the DDL and DML queries to execute the operations given below. Upload the document as a single PDF file. The assessment document should contain the following information:

- Question
- Query
- Output

Exercise – 1

(University Academic Management System)

DEPARTMENT				
Dept_ID	Department_Name	HOD_Name	Office_Location	Budget (₹)
101	Computer Science	Dr. R. Kumar	AB-1 Block	25,00,000
102	Information Technology	Dr. S. Mehta	AB-2 Block	22,00,000
103	Electronics & Communication	Dr. P. Nair	AB-3 Block	26,00,000
104	Mechanical Engineering	Dr. A. Sharma	MB-1 Block	28,00,000
105	Civil Engineering	Dr. D. Singh	CB-1 Block	20,00,000
106	Electrical Engineering	Dr. K. Rao	EB-2 Block	24,00,000
107	Biotechnology	Dr. M. Joseph	BT Block	21,00,000
108	Artificial Intelligence	Dr. N. Das	AI Block	30,00,000
109	Data Science	Dr. T. Patel	DS Block	32,00,000
110	Mathematics	Dr. G. Roy	Science Block	18,00,000

FACULTY					
Faculty_ID	Faculty_Name	Designation	Dept_ID	DOJ	Salary
F101	Dr. Anita Sharma	Professor	101	15-Jul-12	1,45,000.00
F102	Dr. Vivek Rao	Professor	102	11-Jan-14	1,42,000.00
F103	Dr. Rahul Das	Associate Professor	103	20-Jun-16	1,18,000.00
F104	Dr. Priya Nair	Assistant Professor	104	18-May-19	82,000.00
F105	Dr. Arvind Kumar	Professor	105	14-Aug-13	1,38,000.00
F106	Dr. Sneha Gupta	Associate Professor	106	01-Apr-17	1,12,000.00
F107	Dr. Ramesh Patel	Assistant Professor	107	09-Nov-20	78,000.00
F108	Dr. Neha Singh	Professor	108	15-Feb-11	1,55,000.00
F109	Dr. Mohan Roy	Associate Professor	109	30-Sep-18	1,10,000.00
F110	Dr. Kavita Bose	Assistant Professor	110	21-Jul-21	76,000.00

Lab Assessment - 1

STUDENT						
Student_ID	Student_Name	DOB	Gender	Dept_ID	Semester	CGPA
S101	Aditya Sharma	15-May-04	M	101	5	8.9
S102	Harini Subramanian	22-Aug-04	F	102	5	9.12
S103	Arindam Chatterjee	12-Feb-03	M	103	7	8.35
S104	Mehak Kaur	30-Oct-04	F	104	5	8.02
S105	Rohan Deshmukh	18-Jan-03	M	105	7	7.88
S106	Ananya Mishra	09-Sep-04	F	106	5	9.35
S107	Tashi Bhutia	16-Jun-03	M	107	7	8.41
S108	Nandini Reddy	05-Mar-04	F	108	5	9.48
S109	Devansh Patel	28-Apr-04	M	109	5	8.66
S110	Isha Boro	01-Dec-03	F	110	7	8.72
S111	Rahul Nayak	19-Jul-04	M	101	3	7.95
S112	Kavya Menon	11-Nov-04	F	108	3	9.21

COURSE					
Course_ID	Course_Name	Credits	Dept_ID	Faculty_ID	Semester
C101	Database Systems	4	101	F101	5
C102	Operating Systems	4	101	F101	5
C103	Computer Networks	4	102	F102	5
C104	Digital Electronics	3	103	F103	5
C105	Thermodynamics	4	104	F104	5
C106	Structural Analysis	4	105	F105	5
C107	Power Systems	3	106	F106	5
C108	Bioinformatics	3	107	F107	5
C109	Machine Learning	4	108	F108	5
C110	Data Mining	4	109	F109	5
C111	Linear Algebra	3	110	F110	3
C112	Artificial Intelligence	4	108	F108	3

ENROLLMENT				
Enrollment_ID	Student_ID	Course_ID	Enrollment_Date	Grade
E101	S101	C101	15-Jul-25	A
E102	S102	C103	15-Jul-25	A+
E103	S103	C104	15-Jul-25	B+
E104	S104	C105	15-Jul-25	A
E105	S105	C106	15-Jul-25	B
E106	S106	C107	15-Jul-25	A+
E107	S107	C108	15-Jul-25	A
E108	S108	C109	15-Jul-25	O
E109	S109	C110	15-Jul-25	A
E110	S110	C111	15-Jul-25	B+
E111	S111	C102	15-Jul-25	A
E112	S112	C112	15-Jul-25	O

Integrity Constraints to be used are given below:

DEPARTMENT

* Dept_ID Primary Key

FACULTY

* Faculty_ID Primary Key

* Salary > 50000

* Dept_ID Foreign Key

STUDENT

* Student_ID Primary Key

* Gender CHECK('M','F')

* CGPA CHECK(CGPA BETWEEN 0 AND 10)

* Semester CHECK(1-8)

COURSE

* Course_ID Primary Key

* Credits CHECK(2,3,4)

* Faculty_ID Foreign Key

* Dept_ID Foreign Key

ENROLLMENT

* Enrollment_ID Primary Key

* Student_ID Foreign Key

* Course_ID Foreign Key

The schema contains five interrelated tables. All foreign key references are provided below:

Lab Assessment - 1

- FACULTY.Dept_ID → DEPARTMENT.Dept_ID
- STUDENT.Dept_ID → DEPARTMENT.Dept_ID
- COURSE.Dept_ID → DEPARTMENT.Dept_ID
- COURSE.Faculty_ID → FACULTY.Faculty_ID
- ENROLLMENT.Student_ID → STUDENT.Student_ID
- ENROLLMENT.Course_ID → COURSE.Course_ID

- Q1. Create all the above tables using appropriate DDL commands by defining suitable data types and integrity constraints such as Primary Key, Foreign Key, NOT NULL and UNIQUE wherever applicable. After successfully creating the schema, insert all the records provided in the tables above. Describe the table and display the entire table with all inserted records.
- Q2. Retrieve the Student ID, Student Name, Department Name, and Semester of all students who belong to either Computer Science or Information Technology departments and whose CGPA is greater than or equal to 8.20. Display the output in descending order of CGPA.
- Q3. Display the names of faculty members who are handling more than one course along with the total number of courses assigned to each faculty member.
- Q4. Update the semester of all students whose CGPA is greater than 9.00 from Semester 5 to Semester 6. Display the updated records.
- Q5. Add a new column named StudentEmail to the STUDENT table. Populate the email addresses for all students and later modify the column by applying the UNIQUE constraint.
- Q6. Rename the StudentEmail column as UniversityEmail. Verify the changes using DESCRIBE command.
- Q7. Increase the size of FacultyDesignation from VARCHAR2(20) to VARCHAR2(40) using ALTER TABLE.
- Q8. Delete the enrollment record of a student whose enrollment date is earlier than '01-JAN-2025'. Display the updated ENROLLMENT table.
- Q9. Rename the COURSE table as SUBJECT and verify the schema.
- Q10. Remove the UniversityEmail column from STUDENT table and display the modified table structure.

Lab Assessment - 1

Exercise – 2

(University Academic Management System - Enhanced)

Modify the schema created in Exercise-1 by incorporating the following additional relations and appropriate integrity constraints. Use suitable Primary Key, Composite Primary Key, Foreign Key, NOT NULL, CHECK, UNIQUE and DEFAULT constraints wherever applicable.

The following three tables are to be added.

ATTENDANCE				
Student_ID	Course_ID	Total_Classes	Classes_Attended	Attendance_Percentage
S101	C101	40	36	90
S102	C103	42	40	95.24
S103	C104	38	32	84.21
S104	C105	40	34	85
S105	C106	42	30	71.43
S106	C107	41	39	95.12
S107	C108	38	33	86.84
S108	C109	40	39	97.5
S109	C110	39	35	89.74
S110	C111	40	31	77.5

Constraints

- * Composite Primary Key (Student_ID, Course_ID)
- * Student_ID → STUDENT(Student_ID)
- * Course_ID → COURSE(Course_ID)
- * Attendance_Percentage CHECK (Attendance_Percentage BETWEEN 0 AND 100)

EXAM_RESULT					
Student_ID	Course_ID	Internal_Marks	External_Marks	Total_Marks	Result
S101	C101	45	44	89	PASS
S102	C103	48	46	94	PASS
S103	C104	41	39	80	PASS
S104	C105	40	42	82	PASS
S105	C106	33	37	70	PASS
S106	C107	49	47	96	PASS
S107	C108	43	40	83	PASS
S108	C109	50	48	98	PASS
S109	C110	44	43	87	PASS
S110	C111	38	41	79	PASS

Constraints

- * Composite Primary Key(Student_ID, Course_ID)
- * Internal_Marks CHECK (0–50)
- * External_Marks CHECK (0–50)
- * Result DEFAULT 'PASS'

COURSE_PREREQUISITE		
Course_ID	Prerequisite_Course	Minimum_CGPA
C102	C101	7.5
C103	C101	7.8
C109	C101	8.2
C110	C109	8.4
C112	C109	8.5
C107	C104	7
C108	C104	7.2
C106	C105	7.5
C111	C101	6.5
C105	C111	6.8

Constraints

- * Primary Key(Course_ID)
- * Course_ID → COURSE(Course_ID)
- * Prerequisite_Course → COURSE(Course_ID)
- * Minimum_CGPA CHECK(Minimum_CGPA BETWEEN 0 AND 10)

- Q1. Modify the schema created in Exercise-1 by incorporating the new relations ATTENDANCE, COURSE_PREREQUISITE and EXAM_RESULT. Define all necessary integrity constraints including Composite Primary Keys, Foreign Keys, CHECK constraints, DEFAULT values and UNIQUE constraints.
- Q2. Attempt to insert a student record with a Department ID that does not exist. Observe and explain the integrity constraint violated.
- Q3. Attempt to insert duplicate Course IDs into the COURSE table. Explain the error generated.
- Q4. Delete a department that is referenced by multiple students and courses. Explain the behaviour of the database based on the defined foreign key constraints.

- Q5. Modify the foreign key on COURSE table using ALTER TABLE by first dropping the existing constraint and recreating it with ON DELETE SET NULL option.
- Q6. Add CHECK constraint to ensure Attendance Percentage remains between 0 and 100.
- Q7. Modify the Marks column from NUMBER(3) to NUMBER(5,2) without affecting existing records.
- Q8. Alter the STUDENT table by adding a new column named StudentStatus of datatype VARCHAR2(15). Set its default value to 'ACTIVE'. Verify the behaviour by inserting a new student record without specifying the StudentStatus column and display the inserted record.
- Q9. Rename the ATTENDANCE table as STUDENT_ATTENDANCE and verify the changes.
- Q10. Drop the CHECK constraint created in Question 6 and explain its effect by inserting invalid attendance data.



VIT[®]

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)